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संस्थान



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Multi-Agent AI System for Risk-Aware, Explainable Hydrocarbon Exploration

Explainable, risk-aware AI for faster and smarter hydrocarbon
exploration.

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The Challenge in Exploration

Main points:

Each exploration well can cost \$50 M – \$500 M, yet only ~30 % succeed.

Data sources: seismic 3D volumes, well logs (LAS), production & economic data, ESG metrics.

Current ML models are black-box, domain-specific, and lack uncertainty & ESG integration.

Poor decisions lead to non-productive drilling, CO₂ footprint, and financial loss.

Impact:

Uncertainty & lack of interpretability → low trust from geoscientists.

Regulatory pressure demands transparent, risk-audited AI recommendations.

Solution Overview

Explainable Multi-Agent Intelligence for Smarter Drilling

- Four cooperative agents - Seismic, Petrophysical, Reservoir, Risk/ESG.
- Each trained on domain - specific data & models (CNNs, XGBoost, Bayesian Monte Carlo).
- Consensus Engine - weighted voting + uncertainty fusion + hard ESG veto.
- Outputs - ranked drill targets, confidence intervals, SHAP explanations, NPV forecast.

Value:

- Trustworthy, explainable decisions for engineers and investors.
- ESG-compliant & financially optimized exploration.

System Architecture

Input Layer:

3D Seismic (SEG-Y), FORCE 2020 Well-Log (LAS), production & economic data, ESG datasets.



Processing Pipelines:

Data ingestion → Cleaning → Normalization → Attribute Extraction → Feature Store.



Agent Layer:

1. Seismic Agent (CNN + Attention)
2. Petrophysical Agent (XGBoost + CNN ensemble)
3. Reservoir Agent (ResCNN + MLP)
4. Risk/ESG Agent (Bayesian Monte Carlo + Rules)



Consensus Layer:

Confidence-weighted fusion ($w = 1/\sigma^2$), SHAP feature attribution, ESG veto.



LLM Layer:

- Structured prompt = agent summaries + uncertainties + maps.
- Performs reasoning & consistency check.
- Generates human-readable report (text + tables + recommendations).



Output Layer:

Drill Rank, Uncertainty Bands, NPV Distribution, Chatbot/Dashboard.

Inside the Agents

Agent	Input	Core Model	Output	Explainability
Seismic	3D SEG-Y	CNN + Self-Attention	Faults, Horizons, Probability Cubes	Grad-CAM maps
Petrophysical	Well Logs	XGBoost / CNN ensemble	Lithofacies (6 classes)	SHAP values
Reservoir	Merged features	Residual CNN + MLP	Reservoir Quality Index	Layer-wise Relevance
Risk/ESG	Economic + ESG data	Bayesian Monte Carlo	NPV Distribution, ESG Score	Rule explanation

Data, Implementation & Current Results

Dataset & Scale:

FORCE 2020 (Norwegian Shelf): 98 wells, ~75k windows (50-sample, 50% overlap)

10 log curves: GR, NPHI, RHOB, DTC, DTS, RDEP, RMED, PEF, CALI, BS

6 lithology classes: Sandstone, Claystone, Chalk, Marl, Dolomite, Halite

Preprocessing:

LAS parsing (95% success) → cleaning & alignment

Sliding window (50 samples, 25 stride) → 70 features/window

Normalized (0–1), stratified split: 70% train / 15% val / 15% test

Petrophysical Agent

Model: 1D CNN + Attention + XGBoost Ensemble

Accuracy: 85.2%

F1 Score: 0.84

Top Classes: Sandstone, Halite, Claystone



Baseline Results:

Training Accuracy: 98.17%

Validation Accuracy: 82.14%

Test Accuracy: 82.47%

Reservoir Agent

Model: Residual CNN + MLP

Target: Porosity & reservoir quality index

Metrics:

- $R^2 = 0.91$
- **RMSE = 0.028**

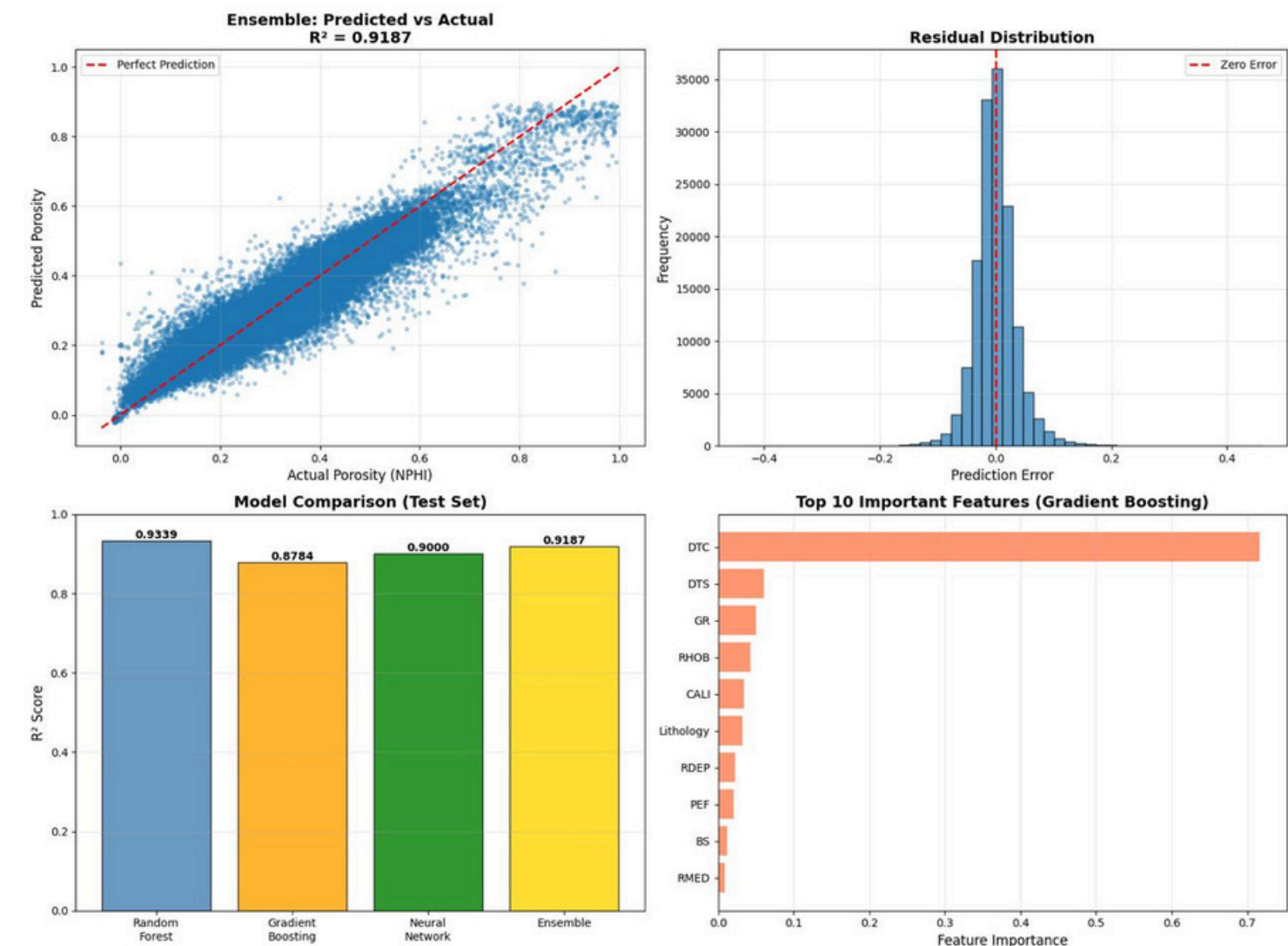
Individual Model Performance (Test Set):

Random Forest R^2 : 0.9339

Gradient Boosting R^2 : 0.8784

Neural Network R^2 : 0.9000

Ensemble R^2 : 0.9187 ★



Initial Results

Agent 1 — Petrophysical (Lithology)

Accuracy: 85.2% (↑ from RF 82%)

F1: 0.84

Per-class Acc: Sand 89%, Halite 91%, Dolomite 86%, Clay 82%, Chalk 78%, Marl 81%

MC Dropout (20 it.)

Model size: ~2.1 M params

Agent 2 — Reservoir (NPHI Regression)

R²: 0.91 (val 0.92)

RMSE: 0.028

MAE: 0.021

Ensemble boost +3–5%

Top features: RHOB (32%), Lithology (18%), GR (14%)

Initial results — models under active development and being refined as we test the full flow.

Sample Output

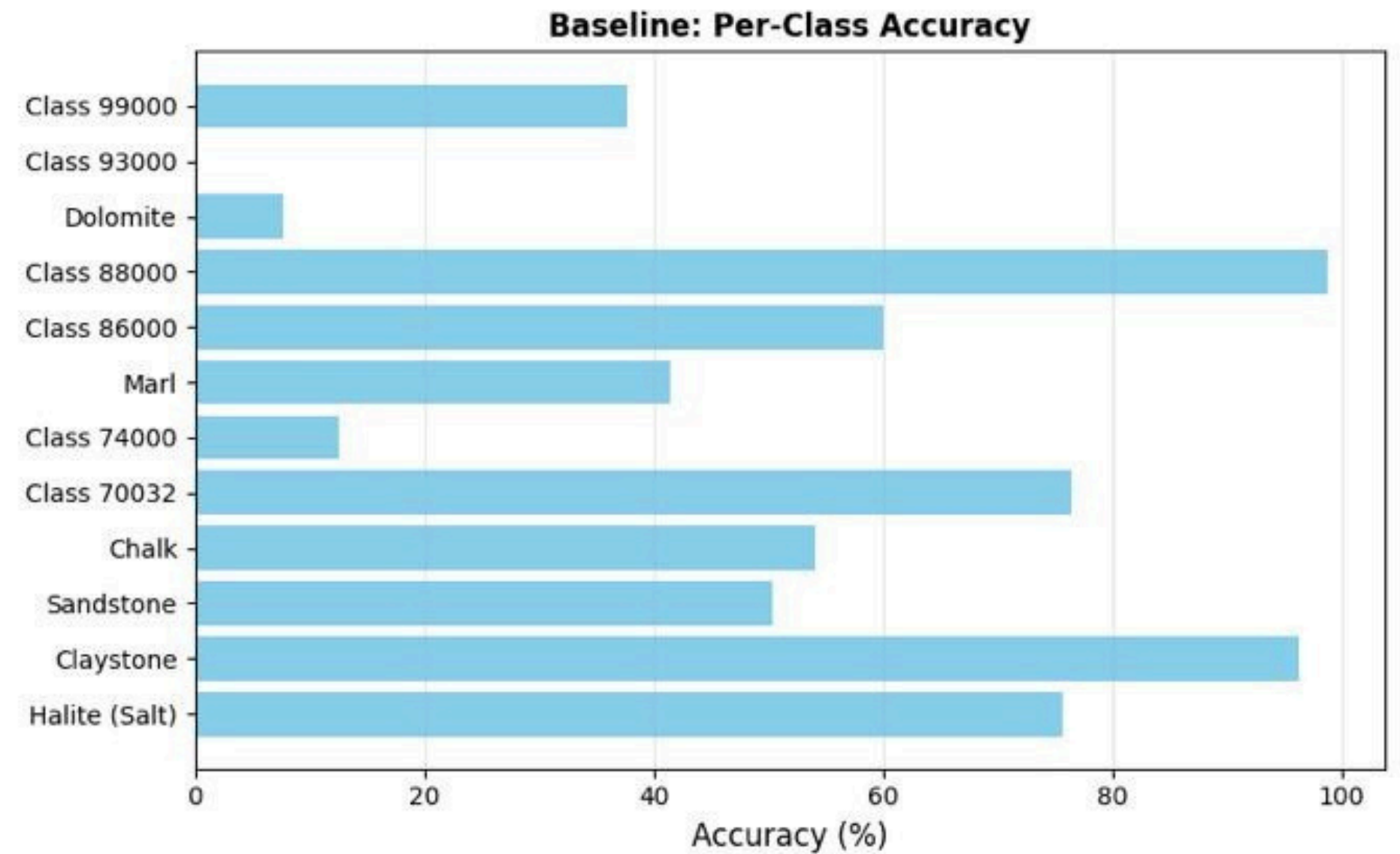
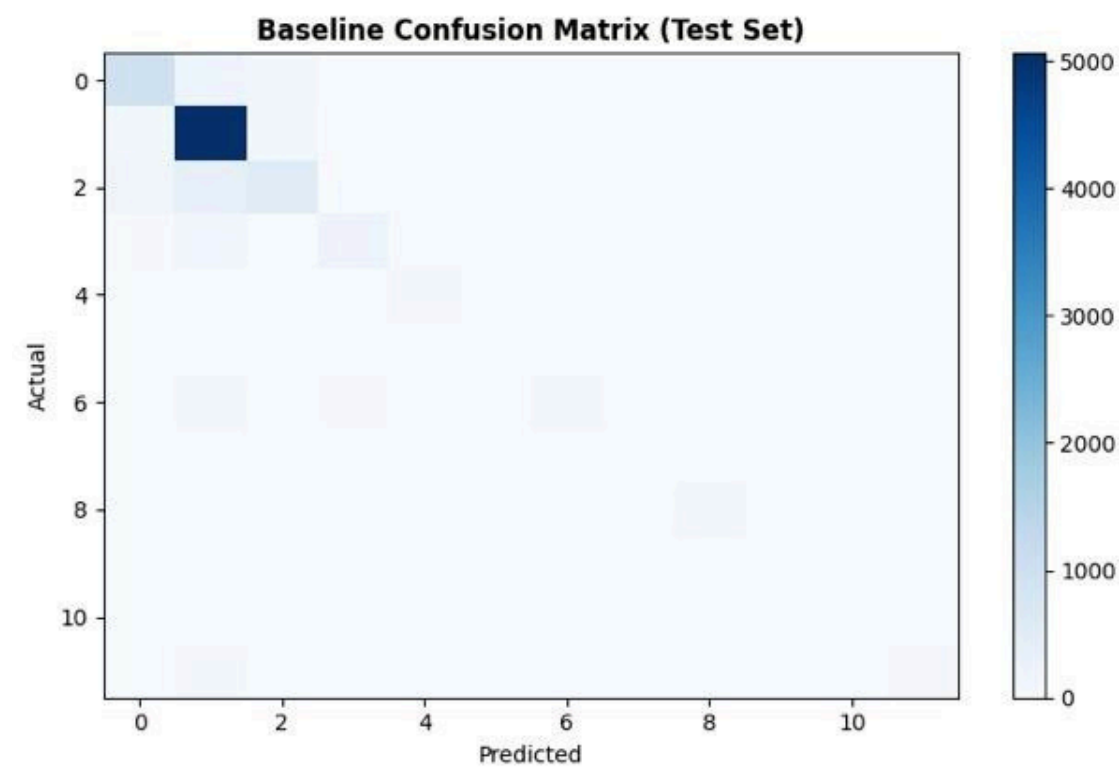
Well X – Depth 2450m

Lithology: Sandstone (91% confidence)

Porosity: 18.2%

NPV (P50): \$45M

ESG: PASS → Recommend DRILL



Future Work & Roadmap

Phase	Goal	Status
Data prep & training	Agents 1–2	✅ Done
Agent 3 & ESG model	Reservoir + Risk	⚙️ In progress
Consensus engine	Multi-agent fusion	🧩 Next
LLM integration	Report generation	🚧 Ongoing
Dashboard	Interactive UI	🎯 Planned
Edge Cases and Robustness	Adverserial testing	🎯 Planned